

Geoxs Biotech – Company Summary

Company Overview

Geoxs Biotech Private Limited, incorporated on 13 September 2022 (CIN: U37100TZ2022PTC040037), is a biotechnology company focused on developing sustainable solutions to address plastic pollution. The company works on enzyme-based plastic degradation technology and algae-derived bioplastic materials, combining biotechnology with sustainable material development.

Innovation

Geoxs Biotech's key innovation is the development and application of microbial enzymes for the degradation of plastic polymers, supported by its existing intellectual property and ongoing development of an extended patent covering its newly developed prototype/machine-based system. The company has also developed a prototype system intended to improve the practical application and scalability of the degradation process. Alongside this, Geoxs is developing algae-based biodegradable plastic materials, creating a complementary approach that addresses both plastic degradation and the development of sustainable alternatives.

Viability

The technology has strong commercial potential because plastic pollution is a major environmental challenge and industries are increasingly seeking sustainable alternatives and waste-management technologies. Geoxs has already progressed beyond the conceptual stage through product development, enzyme technology, patent activity and prototype development. The company can generate revenue through technology licensing, enzyme/product sales, bioplastic materials and partnerships with manufacturers and sustainability-focused organisations. Its planned India–UK business structure also provides

an opportunity to connect Indian R&D and production capabilities with international markets.

Scalability

The technology is designed with scalability in mind. Enzyme production and algae-based material development can initially be carried out at laboratory and pilot scale and subsequently expanded through optimized fermentation, downstream processing and semi-commercial production systems. The developed prototype provides a foundation for further automation and scale-up. Geoxs can progressively expand from laboratory validation → pilot production → commercial production → international market deployment. Collaboration with incubators, research institutions, industrial partners and the UK-based EcoAlgaForm business can further support technology transfer, manufacturing and international commercialization.

Conclusion

Geoxs Biotech aims to build an integrated sustainable biotechnology platform that combines plastic-degrading enzyme technology with algae-based bioplastic development. With existing technology development, prototype progress, intellectual-property expansion and a clear pathway toward pilot-scale production and international commercialization, the company has the potential to develop a scalable solution for the circular plastics economy.